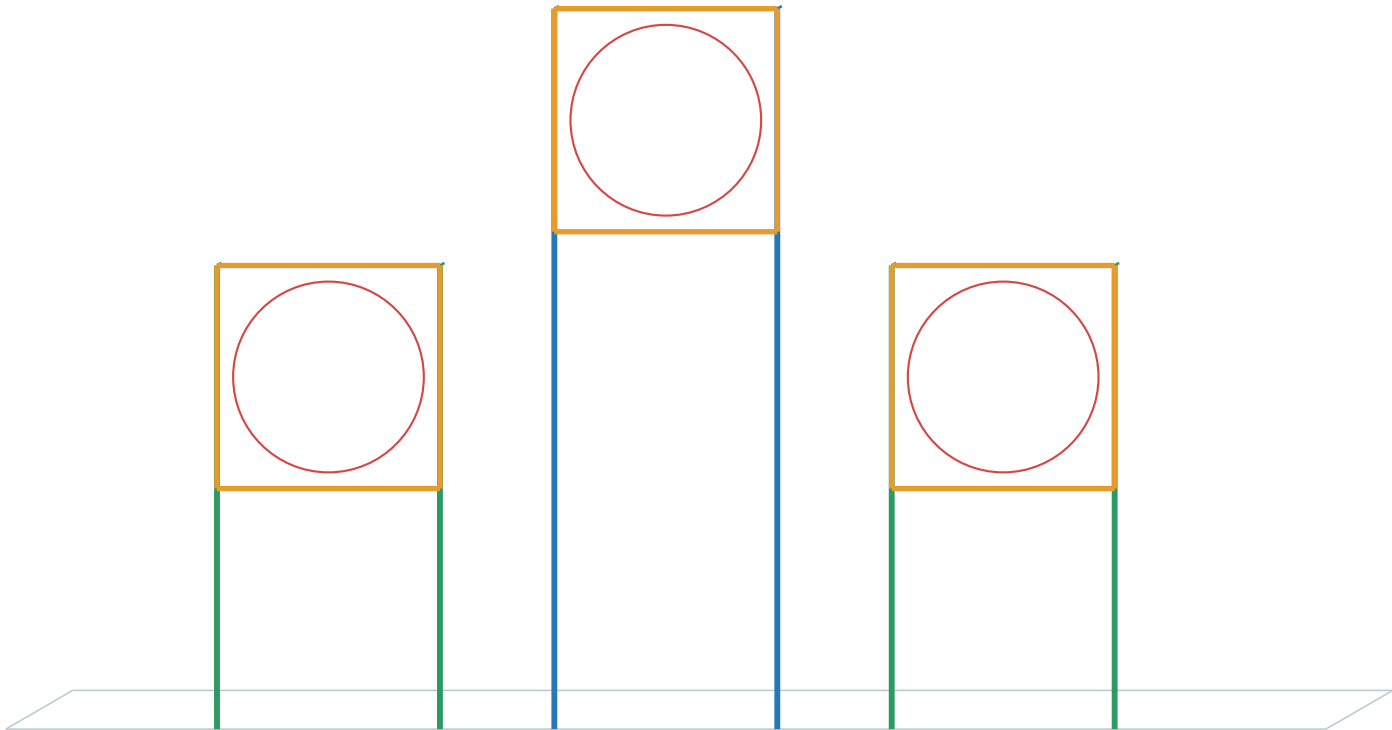


HV-03 EXHAUST-FAN GROUND-SUPPORTED CHASSIS

DESIGN BASIS AND THREE-FAN BAY RELATIONSHIP

3D RELATIONSHIP - PLANAR CHASSIS AT FAN WALL



Fan bodies discharge outside; chassis and bases remain inside wall line.

STRUCTURAL CONCEPT

Planar ground-supported modules avoid obstructing the 750 mm service aisle.

Slotted neoprene-bushed ties restrain out-of-plane movement without carrying fan dead load.

Angles are used for fan mounting and bracing; closed SHS remains at primary posts.

SECTION COLOR KEY

- SHS 50x50x3 - lower-fan posts
- SHS 60x60x3 - upper-fan posts / RHS 50x30x2 links
- L 50x50x5 mounting frames / L 40x40x4 bracing
- 1400x1400 exhaust fan casing

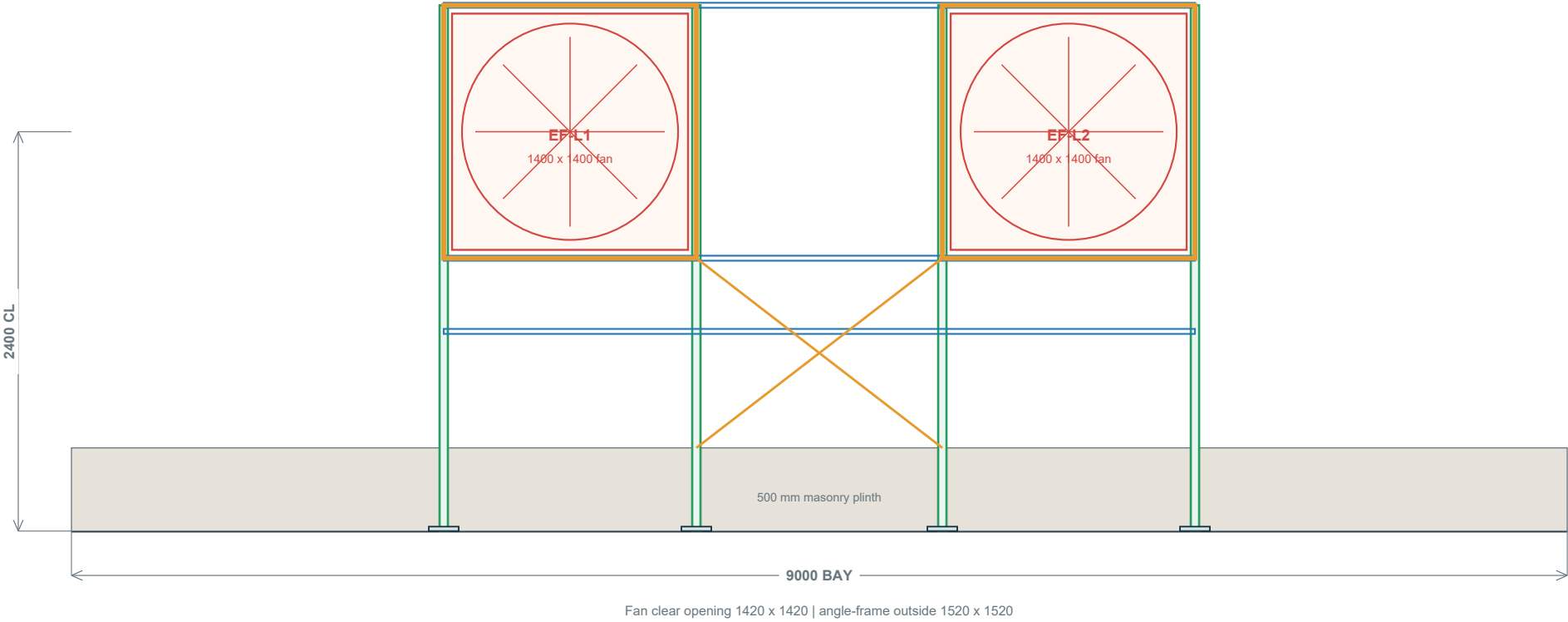
APPROVED DESIGN BASIS

- Production fans: 35; service-greenhouse fans: 3.
- Fan casing: nominal 1400 x 1400; airflow 21,300-24,800 CFM.
- Lower fan centreline: 2400 AFFL; upper fan centreline: 4150 AFFL.
- Chassis carries vertical load directly to concrete; greenhouse only gives lateral restraint.
- Fan body depth: 600 nominal, projecting outside the greenhouse wall.
- Internal fan maintenance aisle remains 750 clear.
- Steel: S235/S275 minimum; all fabricated steel hot-dip galvanized after fabrication.
- Design assumption pending vendor submittal: 150 kg/fan, dynamic factor 2.0.
- Final bolt pattern, casing depth and mass shall be coordinated with approved fan model.

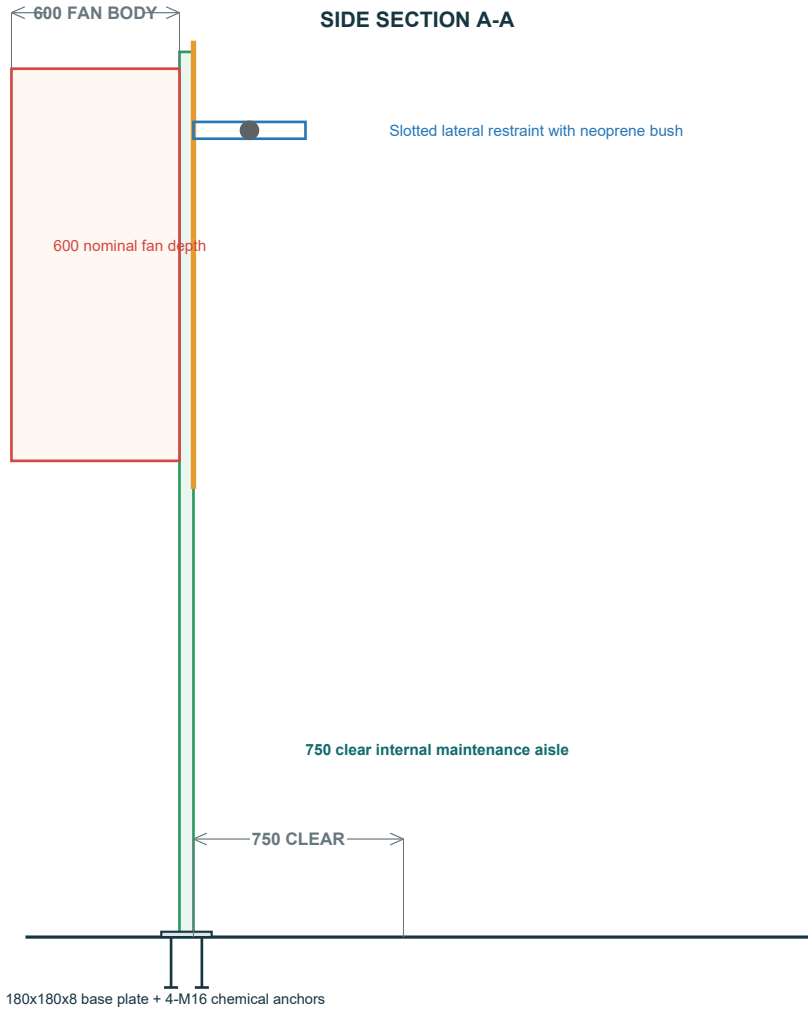
HV-03 EXHAUST-FAN GROUND-SUPPORTED CHASSIS

TYPE A - TWO LOWER FANS PER 9 m BAY

TYPE A - TWO LOWER FANS



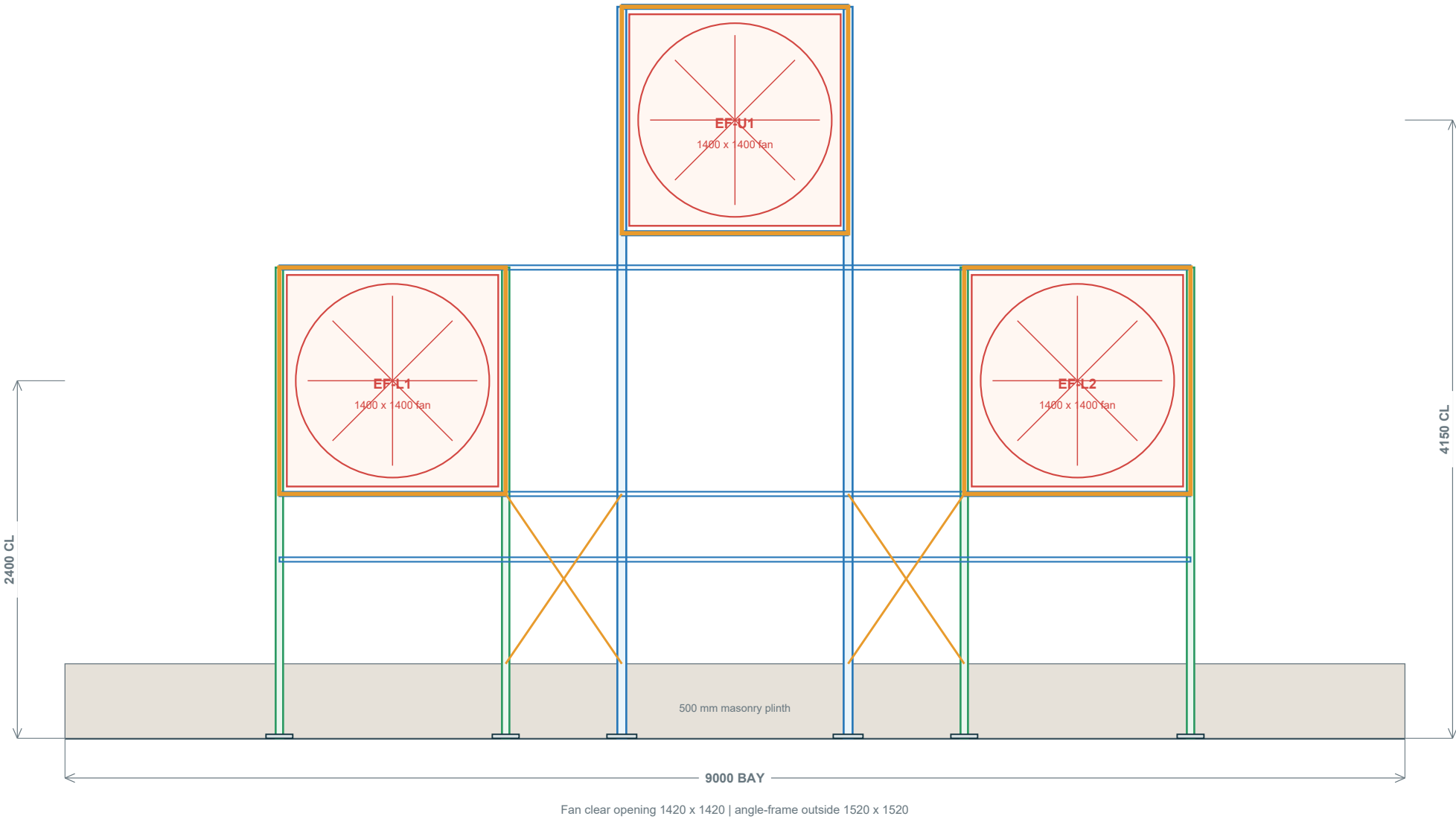
MEMBER	SECTION	QTY	FUNCTION
Lower posts	SHS 50x50x3	4	Ground-supported verticals
Shared rails	RHS 50x30x2	3 levels	Alignment and load sharing
Fan frames	L 50x50x5	2 sets	1420 clear opening
X bracing	L 40x40x4	1 central X	In-plane stability
Base plates	180x180x8	4	4-M16 anchors each
Isolation strip	Neoprene 10 mm	2 sets	Between fan and frame



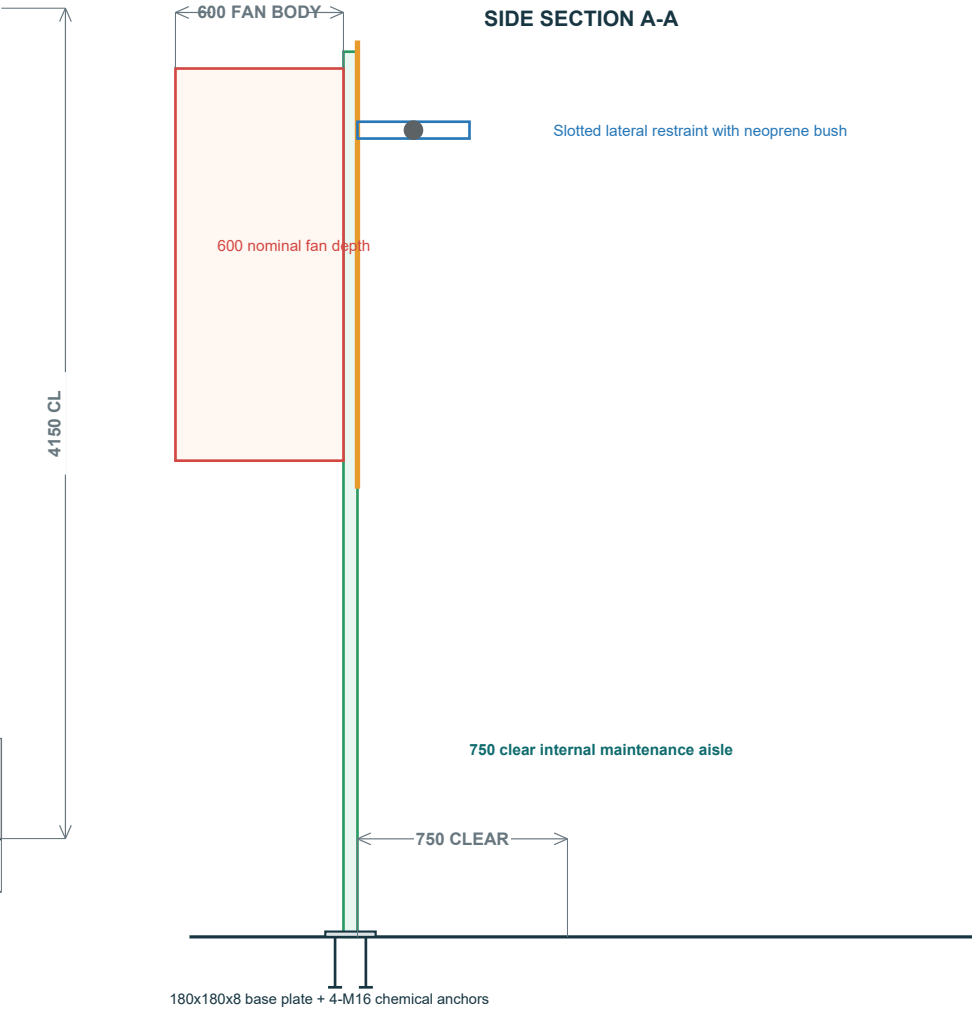
HV-03 EXHAUST-FAN GROUND-SUPPORTED CHASSIS

TYPE B - TWO LOWER AND ONE UPPER FAN PER 9 m BAY

TYPE B - TWO LOWER + ONE UPPER FAN



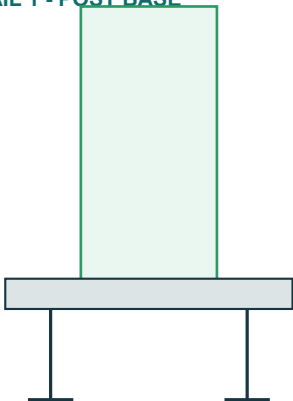
MEMBER	SECTION	QTY	FUNCTION
Lower posts	SHS 50x50x3	4	Lower fan modules
Upper posts	SHS 60x60x3	2	Upper fan module to 4910
Shared rails	RHS 50x30x2	5 levels	Alignment and load sharing
Fan frames	L 50x50x5	3 sets	1420 clear opening
X bracing	L 40x40x4	2 side X	In-plane stability
Base plates	180/200 plates	6	4-M16 anchors each
Isolation strip	Neoprene 10 mm	3 sets	Between fan and frame



HV-03 EXHAUST-FAN GROUND-SUPPORTED CHASSIS

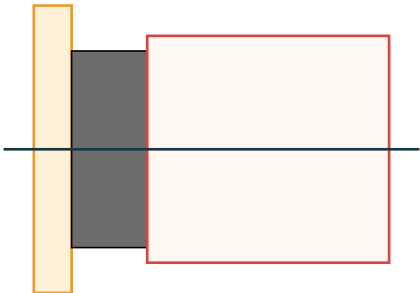
CONNECTION DETAILS, INSTALLATION AND PRELIMINARY TAKEOFF

DETAIL 1 - POST BASE



- SHS post centred on base plate
- 180x180x8 lower / 200x200x10 upper
- 4-M16 chemical anchors; 120 mm min embedment
- 20 mm non-shrink grout; isolate from standing water

DETAIL 2 - FAN TO ANGLE FRAME



- L50x50x5 galvanized mounting angle
- 10 mm neoprene isolation strip
- M12 galvanized bolts + large washers + lock nuts
- Continuous UV-resistant weather seal around casing

ITEM	DESCRIPTION	PROJECT QTY	NOTES
Type A chassis	2 lower fans	7 production bays	Final count by approved bay schedule
Type B chassis	2 lower + 1 upper	7 production bays	Final count by approved bay schedule
Service chassis	Type B	1 service bay	3 service-greenhouse fans
Total fan openings	1400x1400 nominal	38	35 production + 3 service
Hot-dip galvanizing	ISO 1461 basis	All steel	After drilling and welding
Anchors	M16 chemical	By post count	Coordinate with concrete reinforcement
Neoprene isolation	10 mm	38 complete sets	Full perimeter + washers
Flexible weather flashing	GI/EPDM	38 complete sets	Seal fan/plastic interface

MANDATORY HOLD POINTS

- Approve fan shop drawing, actual mass and bolt pattern.
- Survey masonry openings and concrete levels.
- Trial-fit one Type A and one Type B chassis before galvanizing.
- Verify anchor pull-out and edge distances with civil engineer.
- Check free fan shutter movement and 750 mm maintenance aisle.
- Perform vibration check after commissioning.